



Space Propulsion DB-AE 414

Section(s): 01DB;02DB;03DB

Course Information

Academic Terms: Daytona Fall 2024 Term Dates: Aug 26, 2024 - Dec 11, 2024

Credit Hours: 3

Office hours TuTh 11:00-12:00, 3:30-4:30, MoWeFr 3:00-5:00.

Grader has office hours TBD

Instructor Information

Ehsan Roohi Golkhatmi

<https://www.linkedin.com/in/ehsan-roohi-phd-825b47113/>

[Links to an external site.](#)

Email roohie@erau.edu **Office Location** LB227

Catalog Course Description

Introduction to the basic principles of liquid and solid propulsion systems. Flight performance parameters for single and multi-stage vehicles. Thermo-chemistry of the combustion process. Performance enhancements of nuclear rockets and electric propulsion.

Additional Course Description

Prerequisite(s): AE 414 Prerequisite is AE 307 and AE 308.

AE414 focuses on the fundamentals of rocket engines to prepare you for the Astronautics Preliminary AE427 and Detailed Design AE445 courses and the Rocket Propulsion Preliminary AE441 and Detailed AE451 Design courses. It builds on the material you learned in AE308 Compressible Flow. We will not cover Space Mechanics (AE313) or Spacecraft Systems (AE323).

Course Goals

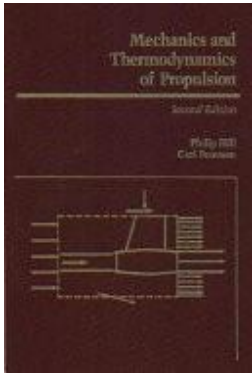
This is a required course for the astronautics track of the aerospace engineering degree program. It is designed to provide depth in propulsion utilizing analytical methods. This will be accomplished by requiring students to demonstrate proficiency in core topics such as fluids, thermodynamics, chemistry and spacecraft design as well as materials and space environment affects.

Student Learning Outcomes

1. Examine mission requirements including launch, on-orbit and deep-space missions and recommend potential classes of propulsion systems that could complete the mission.
2. Explain common propulsion terms like specific impulse, total impulse, delta v, orbital and escape velocity, engine, motor, throat, area ratio, mixture ratio, turbopump, injector, gas generator, grain, binder, frozen-flow, equilibrium flow, monopropellant, particle-bed reactor, electrothermal, electromagnetic and electrostatic thrusters.
3. Calculate combustion chemistry and the fluid mechanics through a nozzle to include on-design and over and under expansion of nozzle flow.
4. Apply mass and volume trends to size propulsion systems and subcomponents to meet flight vehicle envelopes.
5. Analyze main components of various propulsion systems and describe potential limitations for such components and systems.
6. Demonstrate an appreciation for the vast variety of propulsion options available and the breadth of technical knowledge used in each system.

Prerequisite(s): AE 414 Prerequisite is AE 319 or (AE 307 and AE 308).

Required Course Materials



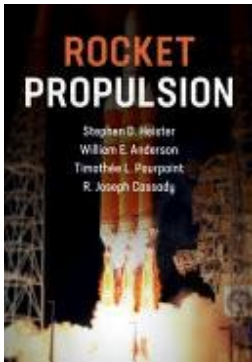
Text/Tool: Mechanics and Thermodynamics of Propulsion

Authors: Philip Graham Hill, Carl R. Peterson

Publisher: Pearson

Publication Date: 1992-01-01

Optional Course Materials



Text/Tool: Rocket Propulsion

ISBN: 9781108422277

Authors: Stephen D. Heister, William E. Anderson, Timothée L.

Pourpoint, R. Joseph Cassady

Publisher: Cambridge University Press

Publication Date: 2019-02-07

Grading

Letter Grade	Percentage
A	90 - 100%
B	80 - 89%
C	70 - 79%
D	60 - 69%
F	< 60%

HW 24%

Exam 1 23%

Exam 2 23%

Final 30%

University Policies

ACADEMIC INTEGRITY

Embry-Riddle is committed to maintaining and upholding academic integrity. Academic integrity violations include cheating, fraud, plagiarism, and double-submissions. More specific definitions of these violations and their consequences are described in the Dean of Students' [Honor Codes and Student Policies](#). To ensure fair and full achievement of degree requirements, students must prepare and present their own work. To show that they have completed their work with academic integrity, students should keep any drafts, notes, calculations, and the like.

Use of Artificial Intelligence

The use of ChatGPT and other artificial intelligence (AI) tools in an educational setting must be guided by ethical principles and academic integrity. Students and faculty may use these tools for research support or aids to enhance their learning and scholarship but should not rely solely on them to produce original work. The University recognizes the potential benefits of AI but acknowledges the potential risks, such as creating unauthentic, inaccurate, biased, or harmful content. Students should follow best practices to ensure authentic scholastic accomplishment and academic integrity, including **avoiding plagiarism or machine ghostwriting**. Ultimately, the University upholds the importance of intellectual honesty and ethical research practices and expects students to act in accordance with these principles when using AI tools.

Course Policies

HW must be turned by the assigned due date.

Tests must be taken on the scheduled date/time unless you have prior permission from the instructor.

HW and tests must be done on engineering paper and be legible.

Student Resources

CAMPUS SAFETY & SECURITY

[Campus Safety & Security](#) officers are on duty 24 hours per day, 365 days a year. We strongly encourage students to report crimes, emergencies, or suspicious conditions to Safety & Security by calling the department's Communications Center at 386-226-6480. In the event of an emergency, call 386-226-SAFE (7233).

CANVAS HELP

When logged in to Canvas, click Help – located at the bottom of the global navigation on the left side. This menu has choices for Canvas Support that you can choose from:

- Search the Canvas Guides
- Canvas Support Hotline at 1-833-334-2831, available 24 hours a day, 7 days a week
- [Chat with Canvas Support](#)

COUNSELING CENTER

The mission of the Counseling Center is to foster student development, growth, and learning. The clinicians provides a calm, safe, supportive environment for students to discuss and explore issues affecting their mental health, wellness, interpersonal relationships, and their academic performance, persistence, and success. Information shared in counseling, including the fact that a student received services, is confidential. Counseling records are separate from academic and administrative records. For students needing more extensive and specialized treatments, referrals to community counselors, psychologists, and psychiatrists are available. The Counseling Center also promotes mental health awareness and wellness through campus-wide activities, programs and presentations.

Wellness posts are available on the department's Facebook, Twitter, and Instagram social media accounts (@eraucounseling). A variety of mental health assessments and resources are available on the department's website, via ERNIE. 24/7 Mental health crisis support via app, telephone or web is also available through MySPP/Telus Health Student Support, which is available through App Store/Google Play.

The Counseling Center is located in the Wellness Center Complex, building 502 and can be reached at 386-226-6035 during normal business hours. For more information about services and hours log into [ERNIE](#) for up-to-date information.

DISABILITY SERVICES

Embry Riddle Aeronautical University facilitates the implementation of reasonable accommodations, for students with disabilities, resulting in barriers to fully accessing University courses, programs and activities. Accommodations are determined through the Disability Support Services (DSS) for individuals with documented qualifying disabilities in accordance with applicable state and federal laws. If you would like to request accommodations due to a disability, please contact the [Disability Support](#)

[Services Office](#) at 386-226-7916 or by email at dbdss@erau.edu. Disability Support Services' administrative office is located in Building #500.

THE CENTER FOR FAITH AND SPIRITUALITY/CHAPLAINS OFFICE

The Chaplains Office and Center of Faith and Spirituality offer confidential, individual meetings with all students to discuss life's challenges; spaces for quiet reflection; and opportunities to learn about other traditions. The [Center for Faith and Spirituality](#) has five prayer and meditation rooms which are open to everyone from 6 a.m. until midnight. There are two chaplains serving the Daytona Beach campus of Embry-Riddle Aeronautical University: Reverend David Keck and Father Karl Bergin (Roman Catholic Chaplain). They work with students of all faiths as well as those from no faith tradition.

FOOD PANTRY

We believe access to food is a human right, accessible to every student. We are committed to educating and distributing aid to our Embry-Riddle community. The mission of the Food Pantry program is to support the academic success of students by providing supplemental food for those who are experiencing food insecurity. Students can apply for support through the [Food Pantry Request form](#). A time will be scheduled with the student once the request is received.

HEALTH SERVICES

Our team of medical professionals, including a University Physician, Licensed Dietitian, Nurses and Insurance Specialist, are dedicated to supporting your health and well-being. Whether you need medical care, have questions or require assistance, please do not hesitate to contact us at [386-226-7917](tel:386-226-7917) or dbhealth@erau.edu. We offer our services on a walk-in or appointment basis, and we are here to support your health needs. [Daytona Beach Health Services](#) clinic is located in Building 500 on the corner of Richard Petty and Clyde Morris Blvd.

HUNT LIBRARY

The Hunt Library is here to help you succeed with finding just the right information resources. For detailed research assistance, please contact [Ask a Librarian](#).

- Website: <https://hunt-answers.erau.edu/ask>
- Email: library@erau.edu
- Phone: 386-226-7656 | 800-678-9428

INTERNATIONAL STUDENT & SCHOLAR SERVICES

International students with questions about I-20s, visas or other related services that assist with the maintenance of their status and immigration compliance should contact [International Student & Scholar Services](#) at 386-226-6579 or dbiss@erau.edu. For other international student related issues and questions such as health insurance, taxes, drivers' licenses, campus work authorization, please contact the International Programs Administration Office at 386-323-8133. Both offices are located on the first floor of New Residence Hall 3.

OFFICE OF DIVERSITY, EQUITY AND INCLUSION

The mission of the Office of Diversity and Inclusion is to advance the campus community's understanding, commitment, and respect for diversity, equity, and inclusion. Through providing and/or supporting education, mentoring, programming, advocacy, and outreach the office fosters an environment that is both beneficial and supportive for all students, faculty, and staff. To learn more about the office, campus offerings and resources, visit the [Office of Diversity and Inclusion ERNIE page](#). Through reviewing the ERNIE page, you can learn ways our office and campus partners strive to create an institutional culture where diversity, equity, and inclusion are ingrained in our community. For additional information please contact diversit@erau.edu or huntk10@erau.edu. Both the Office of Diversity and Inclusion and the Diversity and Inclusion Lounge are located on the first floor of the New Residence Hall 3 within the International Programs Suite.

STUDENT ATHLETE SERVICES

For student-athletes participating in a full schedule of practice, school, matches/games/meets, know that I am aware of the difficult schedule. Please use your available tutors and academic assistance as needed and offered through the Eagle Study Connection.

Faculty Athletic Representative (FAR)

James J. Pembridge, PhD

Email: pembridj@erau.edu

Phone: 386-226-7097

STUDENT GOVERNMENT

[Student Government Association \(SGA\)](#) proudly offers a variety of services to improve student life on campus. Through its three branches, SGA can fund student organizations, advocate for students, and carry out multiple projects. Besides giving students a voice on campus, the SGA also has four divisions including: *The Avion*, Touch-N-Go Productions, Emergency Response Team, and WIKD radio station.

TITLE IX

The Title IX Office oversees compliance of Title IX Sexual Harassment in accordance with Federal Regulations, as well as incidents falling under the University Sexual Misconduct policy. Policy violations can include sex discrimination, sexual harassment, or sexual violence, such as rape, sexual assault, relationship / dating violence, sexual misconduct, and stalking.

Anyone may report suspected or known violations to the Title IX Office and may be able to receive supportive measures. Please see the Title IX website for additional information.

DB Title IX Office

Email: dbtitle9@erau.edu

Office: Student Union, Suite 324

Website: dbtix.erau.edu

[Online Complaint Form](#)

UNDERGRADUATE RESEARCH

Embry-Riddle strives to create a culture of knowledge discovery through research. The [Office of Undergraduate Research](#) engages undergraduate students in faculty-mentored research that is both faculty and student-led. Our mission is to provide a diverse set of opportunities for all undergraduate students to enhance their education through engagement in research, inquiry, innovation, and/or other scholarly projects.

VETERANS STUDENT SERVICES

Being a student veteran can result in a variety of complexities that might require accommodations. Complications with VA benefits disbursements, and other unforeseen military-related developments can complicate your academic life. Therefore, please consider making professors aware of your Veteran status and contacting Veterans Student Services.

Veteran Student Services (VSS) facilitates the transition of military-affiliated students from military culture to University life, supports their academic success through informative programming, and assists veterans, active service members, guardsmen, reservists, and military dependents in receiving their military educational benefits. The unit is staffed with qualified school certifying officials who deliver a broad range of services; while providing informal counseling to students using VA educational benefits. In addition, VSS works in collaboration with our [Student Veterans of America chapter](#) and [Faculty-4-Veterans](#), who support the unit in addressing the needs of our military-affiliated students across campus. VSS, which is located in Residence Hall 3, Bldg. 255, can be reached at 386-226-6350 or dbva@erau.edu

WEATHER EMERGENCY STATEMENT

Hurricanes, tornadoes and other natural disasters (such as fires) are a part of life in Florida. In the event a natural disaster threatens our area, everyone at ERAU is expected to monitor voicemail, email, and the local media for any changes to the normal schedule, including evacuation plans. Decisions to close the Daytona Beach campus are typically made sometime in the afternoon on the day before the intended closure. In the event of an emergency during class hours, please listen carefully to directions from your instructor. If it becomes necessary to evacuate the classroom, we will gather at a designated meeting point away from the building and take attendance to ensure everyone is safe and accounted for. As part of the disaster preparedness process, it is strongly suggested that each member of the ERAU community enroll in the RAVE emergency notification system. If you have not done so, please sign up using the link provided on your ERNIE home page.

Tutoring Services

COLLEGE OF ARTS AND SCIENCES TUTORING

Academic Advancement Center

The [Academic Advancement Center](#) (AAC) is your direct connection to academic support, which provides free tutoring and Supplemental Instruction (SI) to all DB students on a walk-in basis (no appointments needed). The AAC provides peer led tutoring for all foundational math, physics, engineering, chemistry, and biology courses. The AAC is located on the first floor of New Residence Hall 3 (NRH3).

Please refer to the [Academic Advancement Center website](https://campusgroups.erau.edu/erauaac/home/) for details and schedules (<https://campusgroups.erau.edu/erauaac/home/>) and consider following the AAC's Instagram account for updates @erauaac.

Fall & Spring Hours of Operation:

- Monday – Thursday: 9:00 AM – 9:00 PM
- Friday: 9:00 AM – 5:00 PM
- Sunday: 4:00 PM – 8:00 PM

Math Tutoring Lab located in NRH3-112 (MA 111, MA 112, MA 143, MA 241, MA 242, and MA 243)

Physics Tutoring Lab located in NRH3-118 (PS 113, PS 117, PS 150, PS 160, PS 250, PS 226, PS 227, and PS 228)

Engineering Tutoring Lab located in NRH3-113 (EGR 115, EGR 120, ES 201, ES 202, ES 204 Dynamics, and AE 201)

Chemistry Tutoring Lab located in NRH3-126 (CHM 110 and CHM 110L)

Biology Tutoring Lab located in COAS 417 (BIO 120, BIO 305, BIO 306, BIO 321, BIO 322, and BIO 405)

General Study Area located in NRH3-119.

This area is available for all students to use as a general study space. Utilizing this room encourages students to collaborate with their peers and form study groups at their convenience.

Computer Lab located in NRH3-117.

This lab is open to all students and is equipped with standard academic university software and free printing.

MAP LAB located in COAS 301.28

The Math Department provides support for ALEKS and MyLab Math Software and tutoring. (MA 06, MA 143, MA 241, MA 242, MA 243, and upper-level math courses.)

Center for Communication and Digital Media

The Center for Communication and Digital Media (CCDM) combines the former Writing Center and the former Digital Studio to offer comprehensive communication tutoring services. The CCDM offers peer tutoring that adapts to the diverse needs of every communicator. We support students with written, spoken, and digital communication, and the CCDM offers access to the Adobe Creative Cloud, Final Cut Pro, professional audio-visual recording and editing equipment, and a variety of other software and hardware.

Appointments can be made here: <https://erau.mywconline.com>. You must create an account in order to schedule an appointment. The CCDM is located in SU 431, inside the Hunt Library.

COLLEGE OF AVIATION TUTORING

The **Aviation Learning Center (ALC)** located in COA 141 provides free tutoring on Aeronautical Science and Flight course related topics. Fall/Spring Hours of Operations: Monday - Friday, 8 a.m. - 8 p.m.; Saturday 10 a.m. - 5 p.m. Summer hours: 10 a.m. - 5 p.m

The ALC offers access to ten Advanced Aviation Training Devices (AATDs) that are able to simulate both the Cessna 172S and the DA-42 VI, including the Garmin G-1000 avionics, and is equipped with training aids including aircraft components, cut-away instruments, cockpit procedure trainers, Flight Management Systems trainers, reference literature, and much more.

Weather Tutoring Lab is open Monday through Friday from 9:00 AM – 5:00 PM and staffed with both virtual and face-to-face tutors who can help you with many meteorology, math and physics courses. The weather lab is located in COA 356.

COLLEGE OF ENGINEERING TUTORING

Provides tutoring for upper-level 300 & 400 level engineering courses (EGR, ES, AE, CIV, CS, CEC, EE, SE, SYS, and ME). The tutoring lab is located in COE LB 332.

COLLEGE OF BUSINESS TUTORING

COB offers tutoring in Accounting, Management, Economics, Business Writing, Product Management, Finance, Marketing, and Business Analytics. Tutoring is located in COB 271 from Monday to Thursday from 10:00 AM – 5:00 PM in the following courses.

Course Schedule

Week	Dates	Module	HW assign	HW due	Chapters	Topic
1	8/26/24- 8/30/24	1	1		Syllabus 10.1-10.3	Course introduction Rocket performance Rocket performance
2	9/4/24-9/6/24	1	2	1	10.4 10.4	Rocket performance Rocket performance
3	9/9/24- 9/13/24	2	3	2	2.1-2.3, 3.1- 3.3 3.7 3.7	Review of thermodynamics and isentropic flow Review of shocks, expansion fans and nozzle flow Review of shocks, expansion fans and nozzle flow
4	9/16/24- 9/20/24	3	4	3	11.1 11.2 11.2	Rocket thrust chambers Rocket thrust chambers Rocket thrust chambers
5	9/23/24- 9/27/24	3 Review Test 1	5	4	11.2	Rocket thrust chambers Review Test1
6	9/30/24- 10/04/24	4 5		5	Notes Notes 12.1-12.2	Combustion, equilibrium, CEA program Combustion, equilibrium, CEA program Liquid Rockets
7	10/07/24- 10/13/24	5			12.3	Liquid Rockets

					12.4	Liquid Rockets
					12.5	Liquid Rockets
8	10/14/24- 10/18/24	5 6			12.5 12.6 12.6	Liquid Rockets, engine cycles Solid Rockets Solid Rockets
9	10/21/24- 10/25/24	6			12.7 12.7 12.7	Solid Rockets Solid Rockets Solid Rockets
10	10/28/24- 11/1/24	Review Test 2 7				Review Test 2 Nozzles
11	11/4/24- 11/8/24	7 8			11.3, Notes 11.3, Notes Notes	Nozzles Nozzles Hybrid Rockets
12	11/13/24- 15/15/24	8			Notes Notes	Hybrid Rockets Hybrid Rockets
13	11/18/24- 15/22/24	6 9			Notes 14.1,14.2 14.1,14.2	Hybrid Rockets Electric Rockets Electric Rockets
14	11/25/24	9			14.3,14.4	Electric Rockets
15	12/2/24- 12/6/24	9			14.3,14.4 14.5 14.6,14.7	Electric Rockets Electric Rockets Electric Rockets
16	12/9/24- 12/11/24	Review Final				Review Final